- The binding energy of nucleus $B.E = \left(Zm_p + Nm_n - M\right)c^2$
- The average binding energy per nucleon is maximum for iron which is 8.8 MeV.
- Alpha decay: ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$
- β^- decay: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^- + \bar{\nu}$
- β^+ decay: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + e^+ + \nu$
- Gamma emission: ${}^{A}_{Z}X^* \rightarrow {}^{A}_{Z}X + \gamma$
- Law of radioactive decay: $N = N_0\, e^{-\lambda t}$
- In general, after n half lives, the number of nuclei left undecayed is $N = \left(\frac{1}{2}\right)^n N_0$
- The relation between half-life and decay constant is $T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.6931}{\lambda}$.
- Mean lite $\tau = \frac{1}{\lambda}$; $T_{1/2} = \frac{0.6931}{\lambda} = 0.6931\tau$.
- If a heavier nucleus decays into lighter nuclei, it is called nuclear fission
- If two lighter nuclei fuse to form heavier nucleus, it is called nuclear fusion
- In nuclear reactors, the nuclear chain reaction is controlled. In stars, the energy generation is through nuclear fusion.

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